

SMART

GYPSUM
BOARD

The Smarter Way to Build

UNITED GYPSUM®

THE GYPSUM BOARD FABRICATION GUIDE

UNITED GYPSUM®



Gypsum Board Products

United Gypsum prides itself in providing fast, reliable and convenient solutions for your construction needs.

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Certified ISO 9001 and ISO 14001



TABLE OF CONTENTS

Standard Gypsum Board	01
Moisture Resistant Gypsum Board	03
Fire Resistant Gypsum Board	05
Heat Resistant Gypsum Board	07
Suspended Ceiling System	09
Furring Ceiling System	11
Drywall Partition System	13
Why Choose Drywalls?	20
Refining the Constructed Environment	21
Wall Lining	22

SMART GYPSUM BOARDS

Smart Gypsum Board is primarily composed of gypsum and contains an incombustible core while covered with extra tough paper on both sides which provides its strength and durability. The manufacturing of Smart Gypsum Board is aimed at meeting the specific demands of customers while complying with ASTM standards tested on E 119, C471, C472, C473 & C1396. Its development through precision engineering of the highest quality materials makes it a quality-assured product.

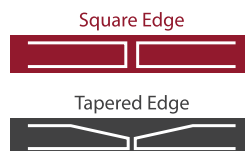
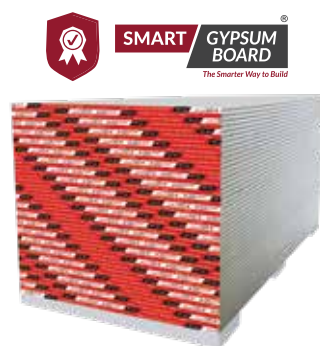
Smart Gypsum Board comes in a comprehensive range of high quality boards, utilizing the best material coupled with advanced engineering to match specific customer requirements:

STANDARD BOARD

Smart Standard Gypsum Board is suitable for interior walls, partitions and ceilings which require no special protection from fire or moisture.

FEATURES

- Convenient and Lightweight
- Quick and Efficient Wall Renovation
- Immaculate and Flawless Surface
- Low Thermal Conductivity and Fire Resistant
- Effortless Setup of Plumbing and Electrical Circuitry



PRODUCT SPECIFICATION

Thickness	Metric Size (mm)	Imperial Size (mm)	Weight (approx.) kg/sheet
7 mm	1200x2400	1220x2440	14
9 mm	1200x2400	1220x2440	19
12 mm	1200x2400	1220x2440	25
15 mm	1200x2400	1220x2440	30

SMART GYPSUM BOARD (STANDARD)

STANDARDS AND TESTS: ASTM C1396, C471, C472, C473

S/No	Parameter	Results
1	Free Water Content	0.23%
2	Weight	8.047 kg/m ²
3	Thermal Conductivity	0.0475 W/mK
4	Density	669 kg/m ³
5	Impact Resistance	0.78 J
6	Fire Resistance	Pass - Non-Combustibility (Thermally Decomposed ≤ 50% @ 750°C x 1 hour)
7	Flame Spread Classification	Flame Spread Index-0.00051 ft. min. (Without Smoke)
8	Breaking Strength Horizontal Vertical	564 N (126.9 lb F) 523 N (117.7 lb F)
9	Compressive Strength Horizontal Vertical	4.6 MPa (667 psi) 4.5 MPa (653 psi)
10	Flexural Strength Horizontal Vertical	4.95 MPa (718 psi) 4.37 MPa (634 psi)
11	Nail Pull Resistance Screw	277 N (62.3 lb F)
12	End Hardness	99.6 N (22.41 lb F)
13	Core Hardness	101 N (22.7 lb F)
14	Edge Hardness	87.3 N (19.6 lb F)



MOISTURE RESISTANT

Smart Moisture Resistant Gypsum Board is an ideal choice for walls, partitions and ceilings that need additional protection from moisture. This special board consists of a blend of wax and a silicone compound enclosed in tough green paper on the surfaces which strengthens walls, partitions and ceilings to endure moist and damp surroundings.



INSTALLATION

To be applied with United Gypsum concealed ceiling metal furring channels and other drywall accessories from the United Gypsum brand.

PRODUCT SPECIFICATION

Thickness	Metric Size (mm)	Imperial Size (mm)	Weight (approx.) kg/sheet
12 mm	1200x1800	1220x1830	20
12 mm	1200x2400	1220x2440	25
15 mm	1200x2400	1220x2440	32

PARAMETERS	RESULTS 12MM THICKNESS
2 Hr Water Absorption (<5%)	2.88%
End Hardness (≥ 49 N)	269 N
Edge Hardness (≥ 49 N)	248 N
Core Hardness (≥ 49 N)	277 N
Thickness (± 0.4 mm)	12.20 mm
Acoustics	Sound Protection STC 35 - 60 dB* when installed with insulation

Square Edge



Tapered Edge



FIRE RESISTANT

SMART Fire Resistant Gypsum Board is an ideal choice for walls, partitions and ceilings that need additional protection from fire. It contains an incombustible fiber added to gypsum; enclosed within a durable, cohesive, compressed pink paper that enhances its strength. This specialized board has been precisely tailored and rigorously tested to meet various fire rating requirements.



INSTALLATION

To be installed with United Gypsum concealed ceiling metal furring channels and other drywall accessories from the United Gypsum brand.

PRODUCT SPECIFICATION

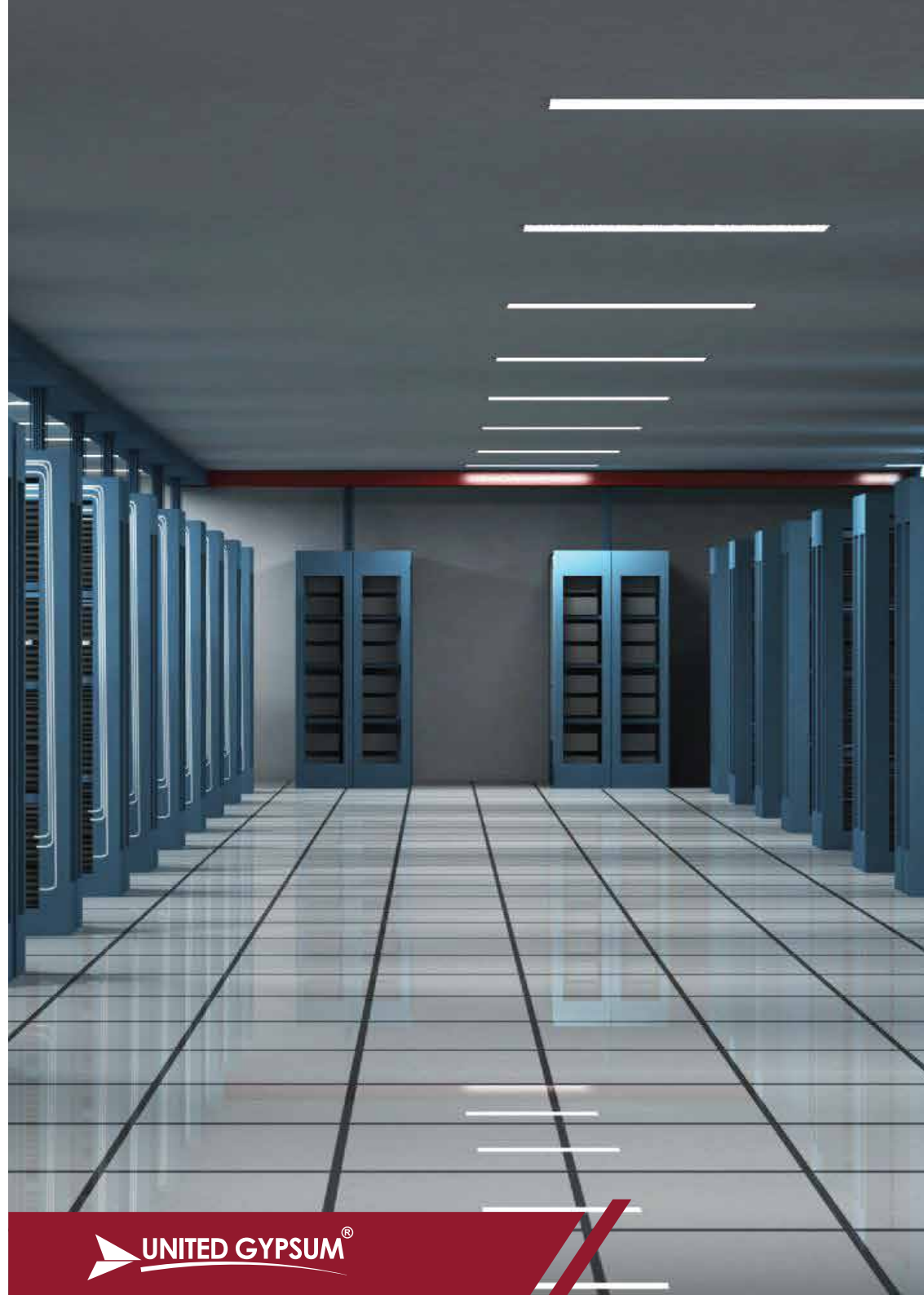
Thickness	Metric Size (mm)	Imperial Size (mm)	Weight (approx.) kg/sheet
12 mm	1200x1800	1220x1830	24
12 mm	1200x2400	1220x2440	30
15 mm	1200x2400	1220x2440	35

PARAMETERS	RESULTS (12MM THICKNESS)
Weight	10.02 kg/m ²
Thermal Conductivity	0.0473 W/mK
Flame Spread Classification	Flame Spread Index - 0.00017 ft. min. (without smoke)
Fire Resistance	Pass - Non-Combustibility (Decomposed ≤23% @ 750°C x 2 hours)
Acoustics	Sound Protection STC 35 - 60 dB* when installed with insulation
Fire Resistance (15mm Thickness)	Pass - Non-Combustibility (Decomposed ≤22% @ 750°C x 2.5 hours)

Square Edge



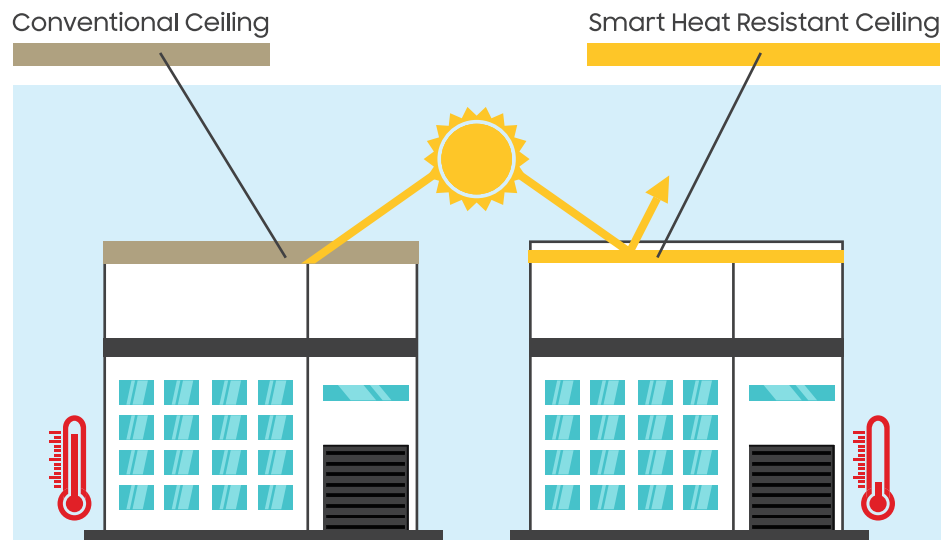
Tapered Edge



HEAT RESISTANT

Smart Heat Resistant Gypsum Board offers effective interior heat insulation for exterior walls and upper-floor ceilings of both commercial and residential buildings that are significantly impacted by heat and sunlight. Beneficial in both hot and cold climates, the board's back liner, attached to a specialized aluminum foil, reflects 95% of thermal radiation, ensuring a comfortable temperature. It also acts as a reliable vapor retarder and prevents moisture penetration in walls and ceilings.

This board is preferred for areas requiring additional heat control and can be incorporated into all Smart Gypsum Boards, including Standard, Fire Resistant, and Moisture Resistant.



SUSPENDED CEILING SYSTEM

Smart Gypsum Board provides great flexibility for an artistic and modish suspended ceiling at an economical rate.

Smart Gypsum Board ceilings provide a much safer environment when compared with the traditional wooden ceilings, with the added capabilities of sound attenuation and fire resistance. With a suspended steel framing garbed in Smart Gypsum Board sheets, these ceilings are ideal not just for a conventional look but for a creative and contemporary look as well.

FEATURES

- Convenient and lightweight.
- Furnishes a fine, immaculate and flawless surface.
- Effortless setup of plumbing and electrical circuitry.
- Remarkable properties of heat, flame and moisture resistance.
- Incomparable flexibility in cutting and flexing to fit curved surfaces to provide an alluring environment.

APPLICATION

Smart Gypsum Board suspended ceilings are ideal for use in virtually every kind of residential or commercial project, including but not limited to private homes, commercial skyscrapers, schools, hospitals, restaurants and factories.



UNITED GYPSUM'S RECOMMENDED FURRING CEILING SYSTEM

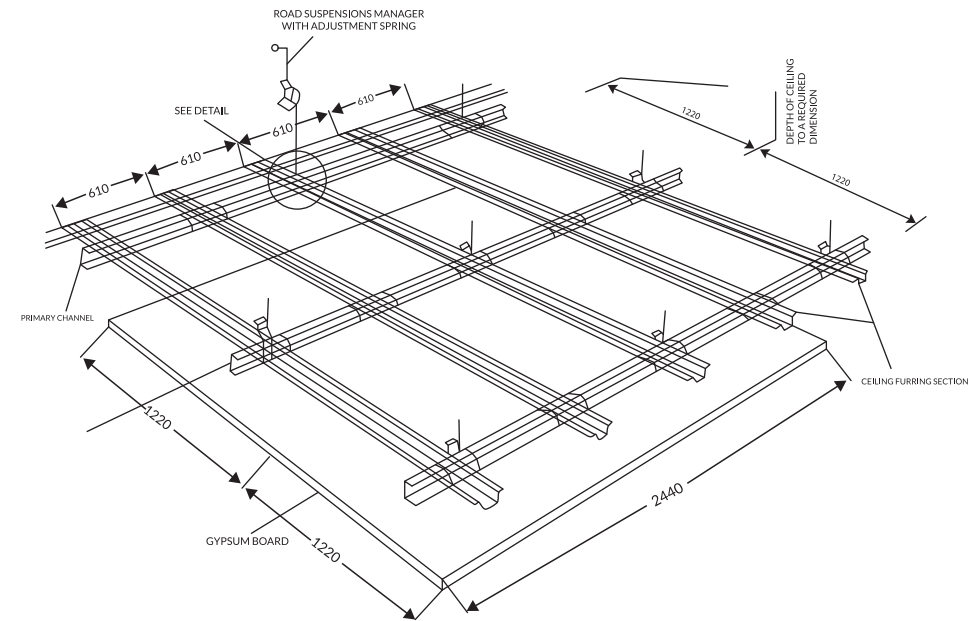
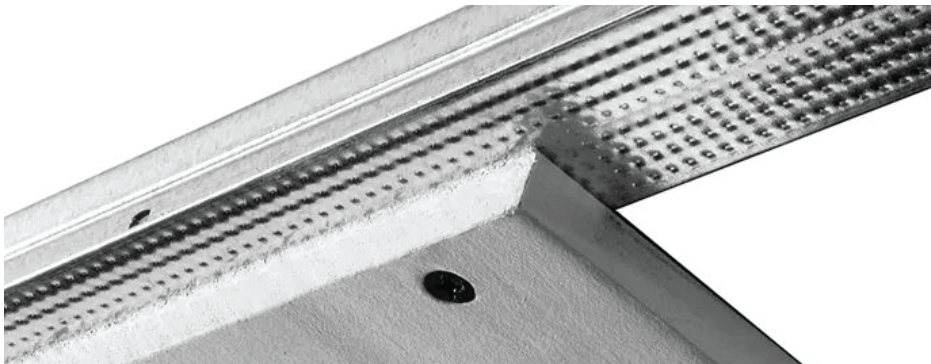
Begin the construction of the furring suspended ceiling system by attaching the 0.5mm thick G.I. wall angle, having equal flanges of 25 mm, to brick or drywall partitions along the perimeter of the ceiling. It can be screw-fixed with the help of nylon sleeves and Smart Screws at an inter-spacing of 610 mm (imperial) or 600 mm (metric).

The primary G.I. channels of 35 mm (0.8mm thick with two flanges of 10 mm each) are suspended from the soffit with a 3 mm suspension wire (GI) and adjustable level clips at an inter-space of 1220 mm (imperial) or 1200 mm (metric). The component is then secured to the soffit with G.I. cleat and steel expansion fasteners.

Once the primary channels have been suspended, 0.5 mm thick furring channels having a knurled web of 35 mm and two flanges of 22 mm each, with lips of 10 mm, are fixed to the primary channels at an inter-space of 610 mm (imperial) or 600 mm (metric).

Next, Smart Gypsum Board conforming to ASTM C1396 of 9 mm or 12 mm thickness are screwed to furring channel with 25 mm Smart Screws at an inter-space of 230 mm. United Gypsum, however, recommends the use of 9 mm gypsum board for suspended ceiling systems since it is not just cost-effective, but also reduces the load on the furring system.

Finally, to cover the joints between the gypsum boards used in the ceiling, utilize Smart Tape (self-adhesive, fiberglass tape) on the tapered or squared edges. For a finer finish, Smart Filler can be used to hide the screw heads before applying paint.



The furring section provides a platform for the gypsum board to be screwed onto via self-tapping drywall screws. The main channel bearer is suspended with the furring top hat section attached with preformed clips.

SPECIFICATION

ASTM C- 1396 Gypsum Boards



DRYWALL PARTITION SYSTEM

A marvel of modern construction techniques, drywall partitions are not just faster to build than traditional masonry walls but are convenient and reliable as well. Compared to wooden partitions, they provide more options, greater fire resistance and are also termite-proof.

A drywall is a panel made of gypsum pressed between two thick sheets of paper. Much like wooden sheets, it is screwed onto a durable frame of galvanized steel to form a partition.

FEATURES

- Sound insulation between 37 to 56 decibels.
- Furnishes a fine, immaculate and flawless surface.
- Ideal for an array of embellishments and coatings.
- Effortless setup of plumbing and electrical circuitry.
- Between half an hour to four hours of fire resistance.
- Economical, lightweight, portable and easily installed with screws.

APPLICATION

Smart Gypsum Board drywall partitions are ideal for use in virtually every kind of residential or commercial project, including but not limited to private homes, commercial skyscrapers, schools, hospitals, restaurants and factories.

WALL SYSTEM ENHANCEMENT

Plumbing, electrical and other services can easily be directed through the cutouts provided in the studs. It is recommended to board one side of the partition first. Furthermore, grommets or isolating strips should be installed in the cutouts to prevent abrasion of the cables. Brackets of G.I. cold-rolled metal sections can be used to mount sockets and switch boxes. To support heavy fixtures like air conditioners or cabinets, additional framing may be required.



**Better
Acoustics**



**Fire
Resistant**



**Robust &
Durable**



**Moisture
Resistant**



**Easy
Installation**

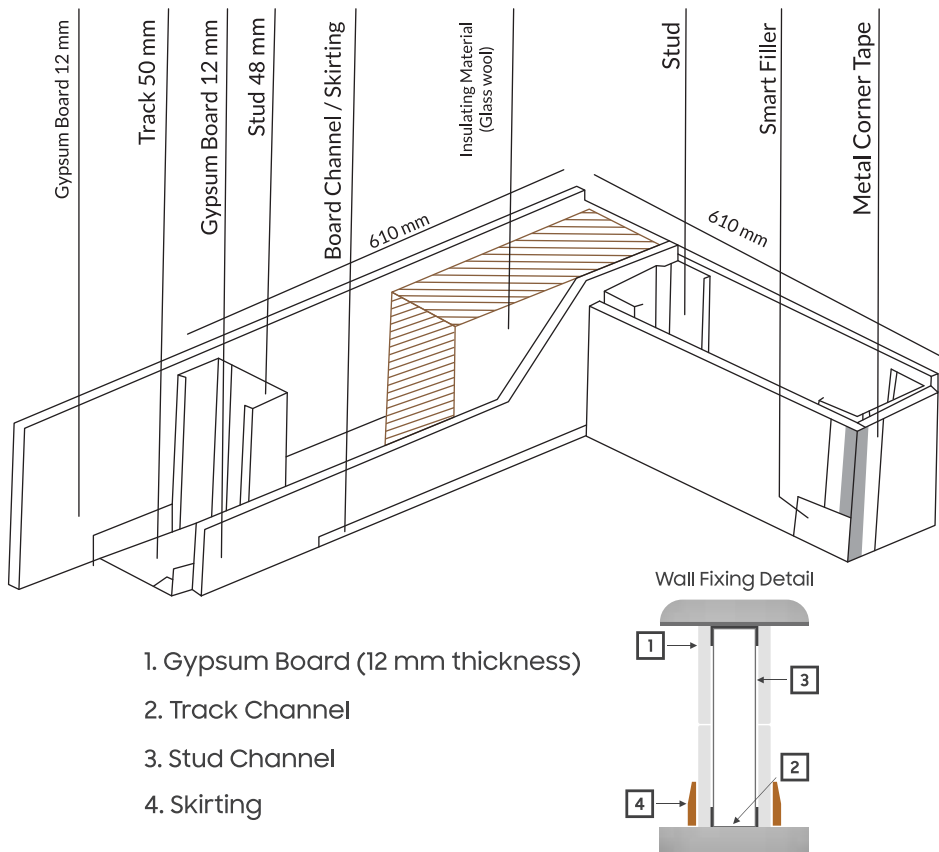


**Minimal
Heat Transfer**



UNITED GYPSUM'S RECOMMENDED SMART GYPSUM BOARD DRYWALL PARTITION SYSTEM

74 mm thick - single layer partition



ESTIMATION BASE FOR 100 SQ. MT.

1220x2440x12 mm Gypsum Board	50 mm Track Channel	48 mm Metal Studs	25 mm Drywall Screws	Jointing Compound	Paper Tape
70 Pcs	84 rmt	174 rmt	2170 Nos	110 kg	320 mts

74 MM METAL STUD DRYWALL PARTITION

A 74mm thick metal stud partition comprises a single layer of Gypsum Board securely attached to G.I. cold-rolled metal sections. This includes a 50mm G.I. track (U channel) for the floor and ceiling, fixed at the head and base (0.5mm thick with equal flanges of 32mm made of G.I.). Vertically fitted G.I. studs measuring 48mm (0.5mm thick with one flange of 34mm and another of 36mm made of G.I.) are friction-fitted within the channel sections to form the framework.

Smart Gypsum Board (conforming to ASTM C1396) with a thickness of 12mm is affixed to the framing, selecting the board type based on environmental conditions. Optionally, 15mm Gypsum Board can be used for better impact resistance. The boards are fixed with Smart Screw™ fine thread drywall screws of 25mm at 305mm (for imperial size boards) or 300mm (for metric size boards) centers on either side of 48mm studs (0.5mm thick with one flange of 34mm and another of 36mm made of G.I.) placed at 610mm (for imperial size boards) or 600mm (for metric size boards) centers in 50mm floor and ceiling channels.

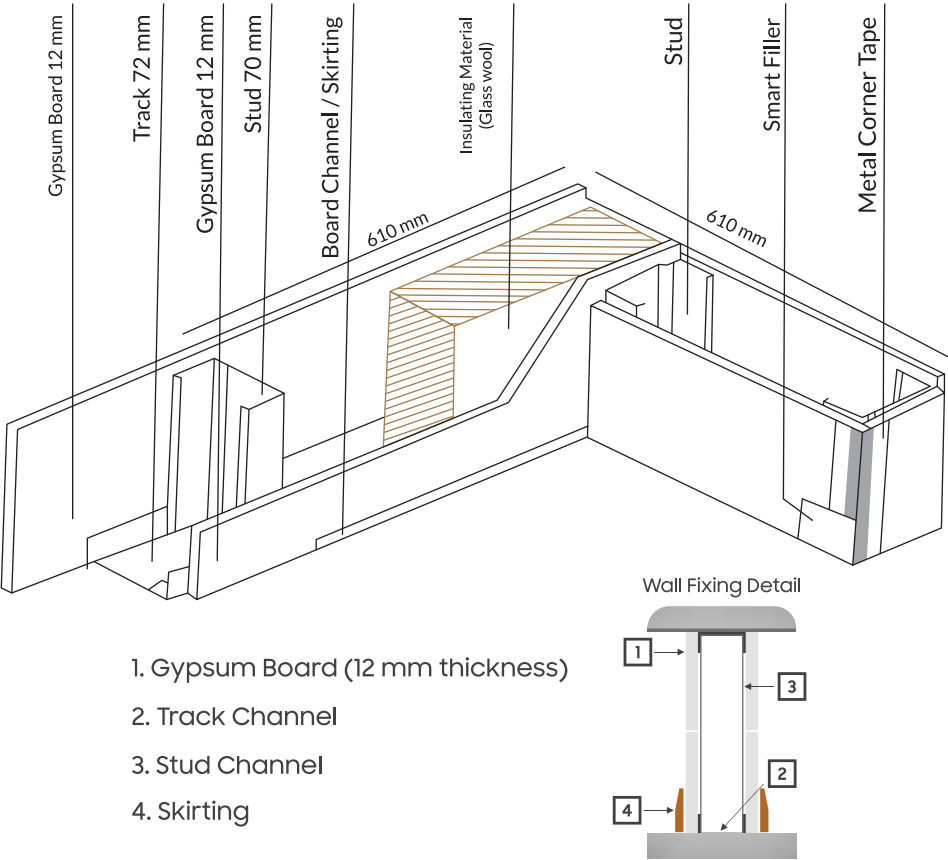
The gypsum boards are installed in a staggered manner to prevent leakage. Finally, square or tapered edge boards, as required, are jointed with Smart Tape™, an ultra-fine self-adhesive fiberglass tape covering joints between the drywall panels. Additionally, Smart Bead™, a Flex Metal corner tape, is used for treating edges and corners to provide excellent impact resistance, preventing edge cracking and chipping.

To finish, joints and corners are treated with Smart Filler™ (Ready-Mix) to hide screw heads and allowed to set/dry before painting (applying three coats of joint filler), leaving it for 24 hours for drying; skimming the board for a better, smooth, and homogeneous look. Exposed compound is ground with abrasive paper, preparing the surface for further decoration such as painting or wallpapering. All surfaces, including applied joint compound, must be thoroughly dry, dust-free, and not glossy before painting.

Drywall partitions constructed using SMART Standard Gypsum Board can provide fire resistance of 60-180 minutes.

UNITED GYPSUM'S RECOMMENDED SMART GYPSUM BOARD DRYWALL PARTITION SYSTEM

96mm thick - single layer partition



ESTIMATION BASE FOR 100 SQ. MT.

1220x2440x12 mm Gypsum Board	72 mm Track Channel	70 mm Metal Studs	25 mm Drywall Screws	Jointing Compound	Paper Tape
56 Pcs	67 rmt	174 rmt	2170 Nos	110 kg	320 mts

96 MM METAL STUD DRYWALL PARTITION

The metal stud partition measuring 96mm thick consists of a single layer of gypsum board securely fastened to G.I. cold-rolled metal sections. This includes a 72mm G.I. track (U Channel) for the floor and ceiling, fixed at the head and base (0.50mm thick with equal flanges of 32mm made of G.I.) Vertically fitted G.I. studs (C Channel) measuring 70mm (0.50mm thick with one flange of 34mm and another of 36mm made of G.I.) form the framework within the channel sections via friction fit.

A layer of tapered edge 12mm thick Smart Gypsum Board (conforming to ASTM C1396) is then affixed to the framing, selecting the board type based on environmental conditions. Optionally, 15mm gypsum board can be used for increased impact resistance. The boards are fixed with Smart Screw™ fine thread drywall screws of 25mm at 305mm (for imperial size boards) or 300mm (for metric size boards) centers on either side of 48mm studs (0.50mm thick with one flange of 34mm and another of 36mm made of G.I.) placed at 610mm (for imperial size boards) or 600mm (for metric size boards) centers in 50mm floor and ceiling channels.

The gypsum boards are to be installed in a staggered manner to prevent leakage. Square or tapered edge boards, as required, are then joined using Smart Tape™ self-adhesive fiberglass tape to cover joints between the drywall panels. Additionally, Smart Bead™ Flex Metal corner tape, combining steel strength with superior bonding compound, is used for treating edges and corners to provide excellent impact resistance, preventing edge cracking and chipping.

To finish, joints and corners are treated with Smart Filler™ (Ready Mix) to conceal screw heads and left to set/dry before painting (applying three coats of joint filler) and allowing 24 hours for drying. Skim the board for a smoother, more uniform appearance. Sand any exposed compound with abrasive paper, ensuring surfaces are thoroughly dry, dust-free, and non-glossy before painting or applying wallpaper for further decoration.

Drywall partitions constructed using SMART Standard Gypsum Board can provide fire resistance of upto 60-180 minutes.



COMPARISON OF DRYWALL PARTITION SYSTEM WITH TRADITIONAL BLOCK MASONRY WALL SYSTEM

Parameters	Masonry Wall System	Gypsum Board Wall
Usage	Durable, tolerant and suitable for external walls	Durable, tolerant and suitable for internal walls
Weight	180 kg / m ²	20 - 75 kg/m ²
Fire Protection	60 minutes of fire resistance	90 - 180 minutes of fire resistance
Acoustic Performance	30 - 40 dB	35 - 80 dB
Speed of Construction	14 m ² /day	40 - 50 m ² /day (3 times faster)
Flexibility of Installation	Requires beam system to support the structure	No beam system required
Loading and Storage	Difficult because heavy	Easy because lighter
Wall Surface	It can only be smooth with skilled labor and cracks can appear	Smooth and crack free surface
Heat Insulation	High heat transmission (K = 0.85 Watt/m ² °C)	Four times lesser heat transmission (K = 0.19 Watt/m ² °C)
Quality of Material	Difficult to control various sources	Standard quality control system

WHY CHOOSE GYPSUM BOARD OVER BRICKS?



REFINING THE CONSTRUCTED ENVIRONMENT

UNITED GYPSUM PROVIDES A RANGE OF SYSTEM SOLUTIONS FOR AN ARRAY OF ENVIRONMENTS TO PROVIDE A HEIGHTENED LEVEL OF SUSTAINABILITY AND SERENITY.

THERMAL ENHANCEMENT

A well-planned and executed thermal insulation can not only help you cut down your building's running expenditures, but also provides a superior level of comfort while helping reduce Carbon emissions. Our Smart Standard Gypsum Board delivers just that, by meeting basic regulatory requirements to the most rigorous, high-performance ones.

ACOUSTIC ENHANCEMENT

United Gypsum Smart Standard Gypsum Board provides acoustic dampening features to enhance the living comfort. An improved acoustic performance delivers various benefits including but not limited to a harmonious environment, greater employee productivity and improved comfort for patients and students.

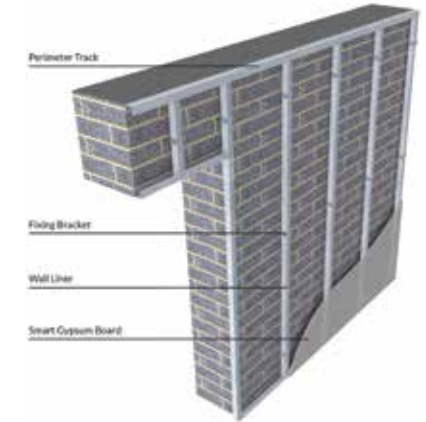


WALL LINING

Smart Standard Board Liner is general-purpose economical and practically self-sustaining metal frame wall lining, ideal for all kinds of non-load-bearing applications in the interior. While these linings can be applied directly on to a plain wall surface, cracked and uneven interior wall surfaces can also be given a fine finish with these linings.

INDEPENDENT METAL FRAME SYSTEM

Where the interior wall surfaces are cracked, uneven or inclined, this self-reliant galvanised steel framework can be attached to the traditional masonry walls to provide a streamlined surface for the Smart Gypsum Boards to be screwed on to before the finishing and decorative touches can be applied.



STEPS FOR INSTALLATION OF WALL LINING

- Mark vertical lines, on the wall to be fixed, at 122 cm intervals from the corners.
- Prepare the drywall adhesive by mixing it with an appropriate volume of water.
- Form a continuous streak of the adhesive along the perimeter of the wall.
- Create three rows of frequent 25 cm long dabs, of a thickness between 5 to 8 cm, along the vertical lines.
- Now place the board near the wall, on a small piece of gypsum board, with the ivory side facing the dabs and firmly affix it with the wall using a straight edge until the board aligns with the lines on the floor and ceiling.
- Cover joints with Smart Tape and apply coats of Smart Filler. Now let the filler set for at least a day.
- Grind the compound with abrasive paper for a smooth finish and the wall is ready for further decoration.

DIRECT BOND SYSTEM

Smart Gypsum Board can be applied directly to the traditional masonry walls as well via the use of a strong adhesive solution, hence providing a quick fix and a flawless surface. However, this method is not recommended for areas exposed to frequent humid conditions.

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- Now place the board near the wall, on a small piece of gypsum board, with the ivory side facing the dabs and firmly affix it with the wall using a straight edge until the board aligns with the lines on the floor and ceiling.
- Cover joints with Smart Tape and apply coats of Smart Filler. Now let the filler set for at least a day.
- Grind the compound with abrasive paper for a smooth finish and the wall is ready for further decoration.

FEATURES

- The most convenient and quickest way to restore or renovate existing walls.
- Excellent heat and sound attenuation properties.
- Independent metal frame covers any seepage damp.
- Furnishes a fine, immaculate and flawless surface.
- Suitable for a multitude of embellishment options.
- Resistant to salt formation.

